

# Optimal control of ODEs

## Exercise sheet

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**Exercise 1** We consider the following problem:

$$\inf_{\substack{T \geq 0 \\ y \in W^{1,\infty}(0,T;\mathbb{R}^2) \\ u \in L^\infty(0,T;\mathbb{R})}} T, \quad \text{subject to :} \begin{cases} \dot{y}_1(t) = u(t) \\ \dot{y}_2(t) = -y_2(t) + u(t) \\ (y_1(0), y_2(0)) = (x_1, x_2) \\ (y_1(T), y_2(T)) = (0, 0) \\ u(t) \in [-1, 1], \end{cases}$$

for a given initial condition  $(x_1, x_2) \in \mathbb{R}^2$ . We will denote by  $y[x, u]$  the solution to the costate equation for a given initial condition  $x$  and a given control  $u$ .

1. Let us assume the existence of a feasible triplet  $(T, y, u)$ . Justify that the problem possesses a solution.
2. Let us fix a solution  $(T, \bar{y}, \bar{u})$  to the problem. Write Pontryagin's principle (we denote by  $(\bar{p}_1, \bar{p}_2)$  the associated costate).
3. Find an explicit expression of  $(\bar{p}_1, \bar{p}_2)$  in function of  $(\bar{p}_1(T), \bar{p}_2(T))$ .
4. Prove that  $\bar{u}$  is piecewise constant, with at most two pieces, and that  $\bar{u}(t) \in \{-1, 1\}$  for a.e.  $t \in [0, T]$ .
5. Find an explicit expression of  $y[x, u]$  for  $u$  constant equal to 1 and for  $u$  constant equal to  $-1$ .
6. Compute the following sets:

$$\begin{aligned} \Gamma_1 &= \{x \in \mathbb{R}^2 \mid \exists T \geq 0, y[x, u = 1](T) = (0, 0)\}, \\ \Gamma_{-1} &= \{x \in \mathbb{R}^2 \mid \exists T \geq 0, y[x, u = -1](T) = (0, 0)\}. \end{aligned}$$

7. On a graph, draw (approximatively)  $\Gamma_1$  and  $\Gamma_{-1}$ . Draw the optimal trajectories of the problem for a set of different initial conditions.

### Exercise 2

- Consider the lunar landing problem introduced in the lecture, with target set  $C = \{(0, y_f)\}$ , where  $y_f > 0$ . Find graphically the set of Pontryagin trajectories (for arbitrary initial conditions).
- Consider the problem introduced in Exercise 1, with target set  $C = \{(0, y_f)\}$ , where  $y_f \in (0, 1)$ . Find graphically the set of Pontryagin trajectories (for arbitrary initial conditions).
- Consider the problem introduced in Exercise 1, with target set  $C = \{(0, y_f)\}$ , where  $y_f \in [1, +\infty)$ . Find graphically the set of Pontryagin trajectories (for arbitrary initial conditions).

**Exercise 3** Consider the following optimal control problem:

$$\inf_{\substack{u \in L^2(0,1) \\ y \in H^1(0,1)}} \int_0^1 \frac{1}{2} y(t)^2 + \frac{1}{6} u(t)^2 dt \quad \text{subject to:} \begin{cases} \dot{y}(t) = y(t) + u(t) \\ y(0) = 1. \end{cases}$$

1. Calculate the pre-Hamiltonian and its derivatives.
2. Justify the existence of a unique solution to the problem.
3. Write the optimality conditions.

4. Let  $(y, u)$  denote the solution, with associated costate  $p$ . Let us set

$$z_1(t) = y(t) - p(t) \quad \text{and} \quad z_2(t) = y(t) + 3p(t).$$

Show that  $z_1$  and  $z_2$  are solutions to independent linear differential equations. Compute  $z_1(t)$  and  $z_2(t)$  in function of  $z_1(0)$  and  $z_2(0)$ .

5. Compute  $y$  and  $p$  in function of  $p(0)$ .

6. Formulate the shooting equation and solve it.

**Exercise 4** We consider the following problem:

$$\inf_{\substack{u \in L^2(0, T; \mathbb{R}^m) \\ y \in H^1(0, T; \mathbb{R}^n)}} \int_0^T \frac{1}{2} \langle y(t), W y(t) \rangle + \langle w, y(t) \rangle + \frac{1}{2} \|u(t)\|^2 dt \quad \text{subject to:} \quad \begin{cases} \dot{y}(t) = A y(t) + B u(t) \\ y(0) = y_0. \end{cases}$$

Data:  $T > 0$ ,  $A \in \mathbb{R}^{n \times n}$ ,  $B \in \mathbb{R}^{n \times m}$ ,  $W \in \mathbb{R}^{n \times n}$ ,  $w \in \mathbb{R}^n$ ,  $y_0$ . The matrix  $W$  is assumed to be symmetric positive semi-definite. Given  $u \in L^2(0, T; \mathbb{R}^m)$ , denote by  $y[u, y_0]$  the solution to the state equation. Let us denote by  $J$  the cost function of the optimal control problem, obtained after elimination of the state variable  $y$ .

We recall two inequalities, useful for this exercise:

$$ab \leq \frac{1}{2}a^2 + \frac{1}{2}b^2 \quad \text{and} \quad (a + b)^2 \leq 2a^2 + 2b^2, \quad \forall a \in \mathbb{R}, \forall b \in \mathbb{R}.$$

We recall that there exists a constant  $M_1 > 0$ , independent of  $y_0$ , such that for all  $u \in L^2(0, T; \mathbb{R}^m)$ ,

$$\|y[u]\|_{L^\infty(0, T; \mathbb{R}^n)}^2 \leq M_1 (\|y_0\|^2 + \|u\|_{L^2(0, T; \mathbb{R}^m)}^2).$$

1. Prove the existence of a constant  $M_2 > 0$ , independent of  $y_0$  and  $w$  such that

$$J(0) \leq M_2 (\|y_0\|^2 + \|w\|^2).$$

2. Let  $u \in L^2(0, T; \mathbb{R}^m)$  and let  $y = y[u, y_0]$ . Prove the existence of a constant  $M_3 > 0$ , depending on  $T$  only, such that

$$J(u) \geq \frac{1}{2} \|u\|_{L^2(0, T; \mathbb{R}^m)}^2 - \frac{M_3}{\varepsilon} \|w\|^2 - \varepsilon M_3 \|y\|_{L^\infty(0, T; \mathbb{R}^n)}^2, \quad \forall \varepsilon > 0.$$

3. Let  $u \in L^2(0, T; \mathbb{R}^m)$  be such that  $J(u) \leq J(0)$ . Prove the existence of  $M_4 > 0$ , independent of  $u$ ,  $y_0$ , and  $w$ , such that

$$\|u\|_{L^2(0, T; \mathbb{R}^m)}^2 \leq M_4 (\|y_0\|^2 + \|w\|^2).$$

4. Prove the existence and uniqueness of a solution to the problem.

**Exercise 5** This exercise is the continuation of the previous one. We denote by  $\bar{u}$  the unique solution to the problem and by  $\bar{y}$  the associated trajectory.

1. Calculate the Fréchet derivative of  $J$ . Find the Riesz representative of  $DJ(\bar{u})$ .

2. Write the optimality system associated with the problem. Denote by  $p$  the adjoint variable. The decoupling technique seen in the lecture can be extended to the new setting. To this purpose, we introduce a function  $E: [0, T] \rightarrow \mathbb{R}^{n \times n}$  and we set:

$$r(t) = p(t) + E(t)\bar{y}(t).$$

Find an expression of  $\dot{r}$  which does not depend on  $p$ . Propose an appropriate Riccati equation for the variable  $E$ , so that the system can be decoupled.

**Exercise 6** We consider the following problem:

$$\inf_{\substack{u \in L^2(0,T;\mathbb{R}^m) \\ y \in H^1(0,T;\mathbb{R}^n)}} \frac{1}{2} \|y(0) - y_0^*\|^2 + \frac{1}{2} \int_0^T \|Cy(t) - f(t)\|^2 + \|u(t)\|^2 dt, \quad \text{subject to: } \dot{y}(t) = Ay(t) + Bu(t).$$

Here  $y_0^* \in \mathbb{R}^n$ ,  $C \in \mathbb{R}^{r \times n}$ ,  $f: [0, T] \rightarrow \mathbb{R}^r$ ,  $A \in \mathbb{R}^{n \times n}$  and  $B \in \mathbb{R}^{n \times m}$ . It can be proved that there exists a unique solution  $(\bar{u}, \bar{y})$ , characterized by the following optimality system:

$$\begin{aligned} \dot{y}(t) &= Ay(t) + Bu(t) \\ B^\top p(t) + u(t) &= 0 \\ -\dot{p}(t) &= A^\top p(t) + C^\top (Cy(t) - f(t)) \\ p(T) &= 0 \\ p(0) &= -(y(0) - y_0^*). \end{aligned}$$

The situation is different from the one seen in the lecture: here, the initial condition of the state is free, which results in an initial condition on the costate.

1. Prove that the above optimality system can be decoupled, as was done in the above exercise. To this purpose, define

$$r(t) = F(t)p(t) + \bar{y}(t),$$

where  $F: [0, T] \rightarrow \mathbb{R}^{n \times n}$  should be defined as the solution to a suitable matrix differential equation.

Consider the situation of a dynamical system  $\hat{y}(t)$ , driven by the equation  $\dot{\hat{y}}(t) = A\hat{y}(t) + Bu(t)$ , where the control  $u$  is an unknown and “small” disturbance, the initial condition is known to be “close” to  $y_0^*$  and  $C\hat{y}(t)$  is known to be “close” to  $f(t)$ . The resolution of the above optimal control problem allows to recover the whole trajectory  $\hat{y}$  (mathematically, one can give a probabilistic sense to “close” and “small” and the optimal trajectory turns out to be the most likely).